BANDARI LAKCHITHA SANKAR REDDY
192571085
CSA1208
COS - AT2

lwo 1'

wo COS-AT2

1. DMA- Based Data transfer.

Introduction :-

Direct memory Access (DMA) is a data transfer technique used to transfer data directly between an I/O device and main memory without requiring the CPU to handle every byte or word of data. DMA is especially usefull when large amounts of data need to be transferred at high speed.

Working of DMA :-

In a DMA - based transfer, the CPU first initializes the DMA controller by providing the Source address, destination address, direction of transfer, and amount of data to be transferred. After initialization, the DMA controller takes control of the required system bus & perform the data. tranfer between the I/O device & memory.

Main Components:-

* DMA controller:- controls & manages the data tranfer.

* I/O Device:- Acts as the source or destination of data.

* Main memory:- Stores the tranffered data.

* CPU:- Initializes the DMA operation & responds to completion interrupts.

Advantages of DMA:-

* Reduces cpu involvement.

* Improves overall system performance.

* Supports high-speed data transfers.

Conclusion:-

→ DMA provides an efficient method for transferring large amounts of data between I/O devices & memory. By reducing cpu involvement & using interrupts to indicate completion.

Vectored Interrupt Handling :-

Introduction:-

* Vectored interrupts are used to efficiently handle interrupts from multiple devices. Each interrupt has a specific vector pointing to it's service routine.

Working :-

-> When an interrupt occurs, the system identifies the device, checks it's priority & excutes the corresponding interrupt service Routine (ISR).

-> After servicing, control returns to the main program.

Advantages:-

* fast interrupt handling.
* Supports multiple devices
* Allow priority - based processing

Conclusion:-

-> Vectored interrupt handling provides an organised & efficient way to manage multiple interrupts.

3. Memory - Mapped I/O & I/O - Mapped I/O

Memory - Mapped I/O:-

* I/O devices are assigned addresses within the memory address space. Normal memory instructions can be used to communicate with devices.

I/O - Mapped I/O:-

* I/O devices use a separate address space & special I/O instructions are used for communication.

Comparision:-

Memory - Mapped I/O	I/O Mapped I/O
* Shares memory address space.	* Uses separate I/O address space.
* Uses normal memory Instructions.	* Uses special I/O Instructions
* more flexible access	* Dedicated I/O addressing.

Conclusion:-

The suitable technique can be selected based on system requirements such as latency & bandwidth.

Lab no:3 Arbitration - PCIe, USB & Thunderbolt.

Introduction:-

Bus arbitration manages access to communication resources when multiple devices request access simultaneously.

Working:-

The system collects device requests, determined priority, grants access to a suitable device, & releases the resource after data transfer

Interfaces:-

* PCIe:- High-Speed interface for internal components and expansion devices.
* USB:- Common interface for External peripherals.
* Thunderbolt:- High- speed interface for demanding peripheral devices.

Conclusion:-

Bus arbitration ensured Efficient and organized communication between multiple devices.

5. Hybrid I/O Communication:-

Introduction:-

A hybrid I/O mechanism uses different communication methods depending on device speed. The assessment specifies polling for low-speed devices & DMA with interrupts for high-speed devices.

Low-Speed devices - Polling:-

* The CPU repeatedly checks the device status until data is available. This is suitable for device with low data rates.

High-Speed Devices - DMA + Interrupts:-

DMA transfers data directly between the device & memory. An interrupt informs the CPU when the transfer is completed.

Advantages:-

* Efficient CPU utilization
* Suitable for different device speeds.
* Faster handling of high-speed data.

Conclusion:- Hybrid I/O provides an efficient communication mechanism by combining polling & DMA with interrupts.